

DATA PROJECT REPORT

Customer Churn Risk Assessment

A predictive analysis of subscription attrition drivers across the Northbridge Retail Group customer base, with a scored risk model and retention recommendations.

PREPARED FOR	PREPARED BY	DATE / VERSION
Northbridge Retail Group	Elegant Data — Analytics Team	July 23, 2026 — v1.0

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Project Scope

This engagement quantifies churn risk for active subscription accounts and identifies the operational levers most likely to reduce voluntary cancellations over the next two quarters. The analysis covers accounts active as of January 1, 2026, and excludes accounts on legacy pricing plans discontinued prior to 2024.

IN SCOPE

What this report covers

- Behavioral and transactional churn signals
- A scored risk model across the active customer base
- Segment-level attrition drivers
- Retention actions ranked by expected impact

OUT OF SCOPE

What this report does not cover

- Pricing strategy and plan design changes
- Support ticket sentiment analysis
- International markets outside North America
- Legacy plan accounts (pre-2024)

Objectives

1. Identify the top five predictors of 90-day churn risk.
2. Deliver a per-account risk score usable by the customer success team.
3. Recommend targeted retention interventions with projected retained-revenue impact.

Tools & Versions Utilized

The stack below was used end to end, from raw event ingestion through model training and delivery.

TOOL	VERSION	ROLE
Python	3.11.7	Data processing & modeling
pandas	2.2.1	Data wrangling
scikit-learn	1.4.0	Risk model (gradient boosted trees)
PostgreSQL	15.4	Warehouse source of truth
dbt	1.7.4	Transformation & testing
Looker	24.2	Stakeholder dashboards
Snowflake	7.36	Feature store

Environment notes

All version numbers reflect the environment pinned at model training time (June 2026). Reproducing results requires the exact dependency set recorded in the project's `requirements.lock` file.

Methodology

The risk model was built as a supervised binary classifier, trained to predict cancellation within a 90-day forward window.

STEP 1 Data extraction 18 months of account, usage and billing events pulled from the warehouse.	STEP 2 Feature engineering 42 behavioral and transactional features constructed at account-week grain.	STEP 3 Model training Gradient boosted trees, validated on a rolling time-split holdout.	STEP 4 Scoring & review Weekly batch scoring, reviewed against realized outcomes.
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Validation approach

Model performance was assessed on a chronological holdout (the most recent eight weeks of data), avoiding leakage from future account behavior. Precision and recall were weighted toward recall, prioritizing the identification of at-risk accounts over avoiding false positives, given the low cost of a retention outreach.

Data Sources & Quality

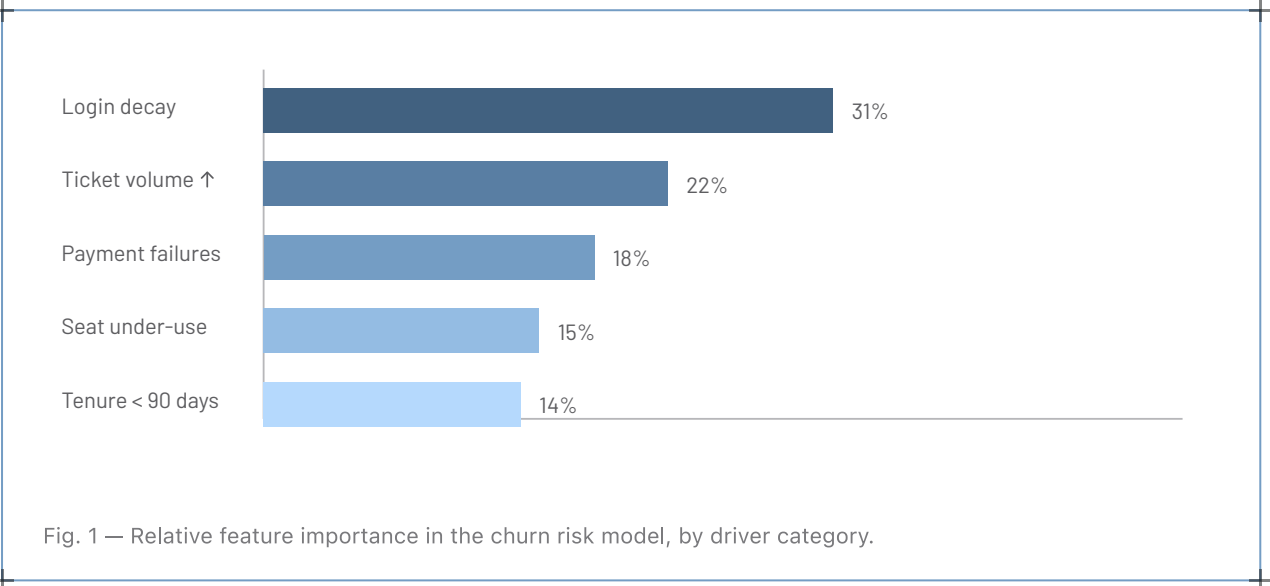
SOURCE	RECORDS	COVERAGE	QUALITY NOTE
Billing system	1.2M rows	100% of active accounts	Complete
Product usage events	84M rows	96% of active accounts	4% gap on mobile-only clients
Support tickets	310K rows	100%	Complete
CRM (account tier, tenure)	62K rows	100%	Complete

Mobile-only usage gaps were imputed using the account's prior 30-day average where available; accounts with no prior history were flagged and excluded from model training but retained in the reporting dataset.

Findings

OVERALL RISK 14.8% of active accounts scored high-risk (>70) for 90-day churn	MODEL LIFT 3.4× over baseline random targeting at equal outreach volume	TOP DRIVER Login decay accounts for 31% of model importance
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Churn risk by driver



Risk by account segment

SEGMENT	ACCOUNTS	HIGH-RISK SHARE	ARR AT RISK
SMB — self-serve	41,200	21.3%	\$3.1M
Mid-market	16,800	11.7%	\$5.6M
Enterprise	3,400	6.2%	\$4.9M
Trial-converted	600	28.4%	\$0.4M

Table 1 — Churn risk concentration and revenue exposure by segment, as of June 2026.

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Recommendations

1

Trigger re-engagement outreach on login decay

Automated outreach when weekly logins drop >50% over a rolling 3-week window.

High impact

2

Proactive dunning for payment failures

Move from single retry to a 3-attempt cadence with in-app payment prompts.

High impact

3

Seat-utilization nudges for mid-market accounts

Quarterly usage reviews for accounts under 40% seat utilization.

Medium impact

4

Extended onboarding for sub-90-day accounts

A guided 60-day onboarding track for the highest-churn tenure band.

Medium impact

Next Steps

ACTION	OWNER	TARGET DATE
Ship weekly risk scores to CS tooling	Elegant Data / Eng	Aug 8, 2026
Pilot login-decay outreach on SMB segment	Northbridge CS	Aug 15, 2026
Update dunning cadence	Northbridge Billing	Aug 22, 2026
Review pilot results, expand to mid-market	Joint	Sep 30, 2026

Appendix – Glossary & Field Definitions

TERM	DEFINITION
Risk score	Model-predicted probability of cancellation within 90 days, scaled 0-100.
High-risk account	An account scoring above 70 on the risk scale.
Login decay	Percentage drop in weekly active logins versus the account's trailing 8-week average.
ARR at risk	Annual recurring revenue held by accounts currently scored high-risk.